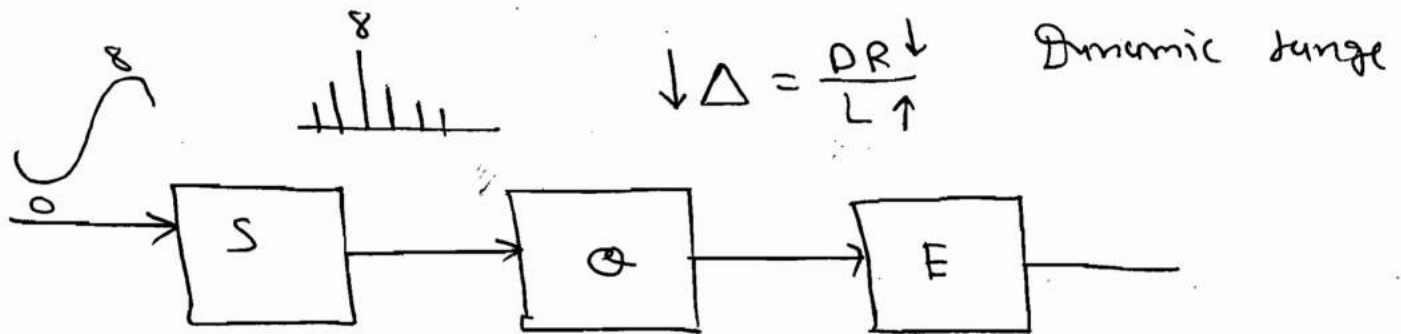


★ DPCM (Differential Pulse Code Modulation):

⇒ DPCM is used to reduce the quantization error without increasing the no. of bits.

⇒ Consider a PCM system and assume that the i/p signal varies from 0 to +8V.



$$n=2 \quad \Delta = \frac{8-0}{4} = 2V \quad [Q_e]_{\max} = 1V$$

$$n=3 \quad \Delta = \frac{8-0}{8} = 1V \quad [Q_e]_{\max} = 0.5V$$

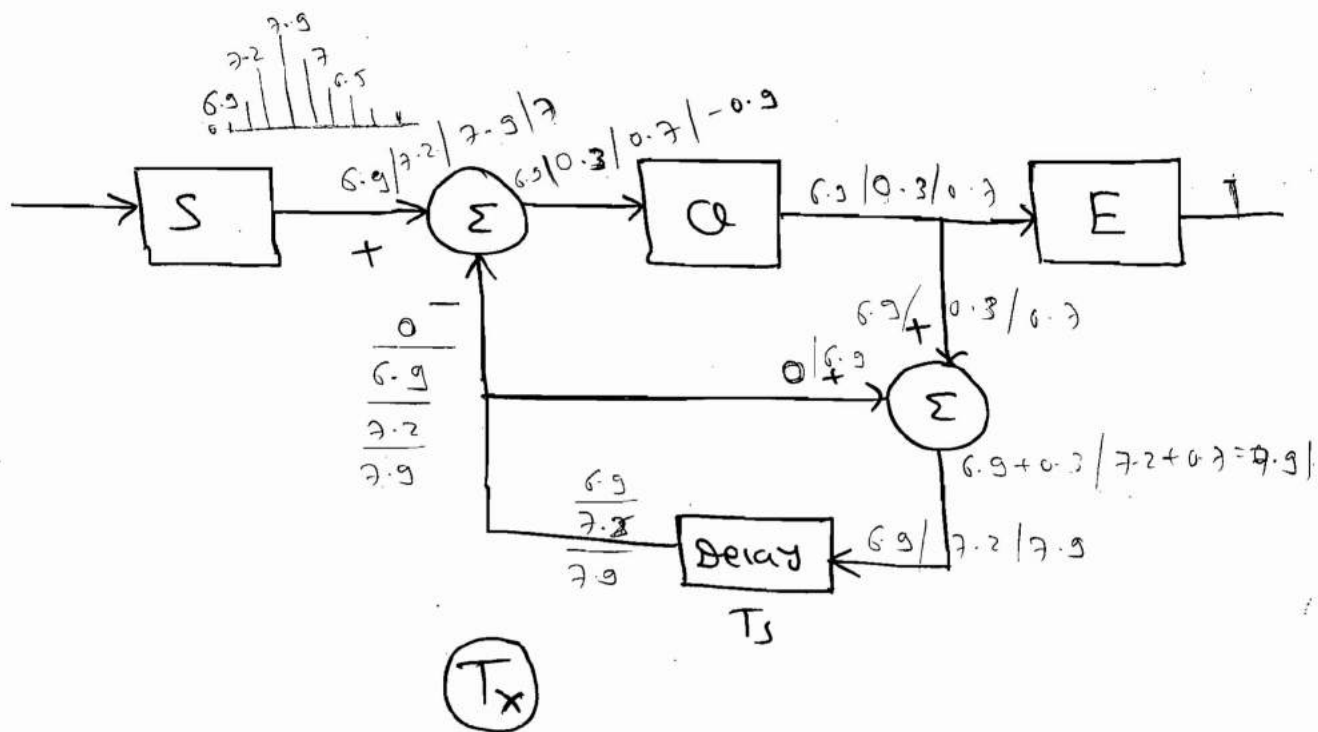
$$n=4 \quad \Delta = \frac{8}{16} = 0.5V \quad [Q_e]_{\max} = 0.25V$$

⇒ Quantization error depends on the step size. In a PCM system, the no. of levels are increased to reduce the step size.

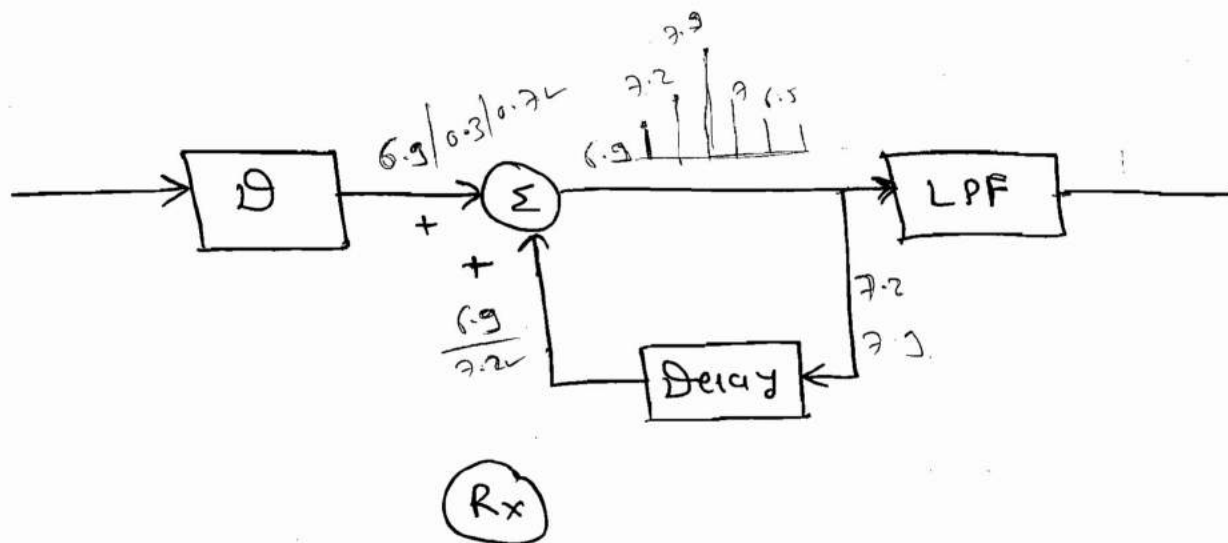
⇒ In DPCM, dynamic range of the quantizer is reduced to reduce the

* Block Diagram of DPCM Tx:

⇒



* Block Diagram of DPCM Rx:



⇒ In a PCM system the samples are applied directly to quantizer. so, the dynamic range is very high.

⇒ In a DPCM system the difference betⁿ ...

dynamic range.

⇒ In a PCM system, the dynamic Range is independent Sampling rate.

⇒ In a dpcm system, the Dynamic Range Varies with the Sampling rate.

⇒ The hardware Complexity of a DPCM System is very high when Compared with PCM.

⇒ The bit rate of the DPCM signal is Same as the PCM.

⇒ In a PCM system, each sample is encoded into n -bits after quantization. This n bits are transmitted through the channel in T_s seconds.

⇒ In DPCM, the difference betⁿ two ~~sets~~ successive Sample is quantized and encoded into n -bits. So, the Bit rate of the DPCM signal is same as the PCM.

